Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS

2020/2021 ACADEMIC YEAR

SECOND YEAR SECOND SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

COURSE CODE: COM 220

COURSE TITLE: SOFTWARE ENGINEERING

MARKING SCHEME

DATE: 15th JUNE 2022 **TIME: 2 P.M.**

INSTRUCTION TO CANDIDATES

- SEE INSIDE

Instructions: Answer ALL Questions in SECTION A and ANY THREE in SECTION B

SECTION A is COMPULSORY (Carries 31 Marks)

QUESTION 1: (COMPULSORY) 16 MARKS

(a) Define the following terms as used in software engineering. (4 marks)

Award one mark for each

i. Software

A set of instructions that are designed to perform a specific task.

ii. Software metrics

Software Metrics provide measures for various aspects of software process and software product.

iii. Risk Management

Risk management is concerned with identifying risks and drawing up plans to minimize their effect on a project.

iv. Prototype

Used when requirements are not well understood and serves as a mechanism for identifying software requirements.

Focuses on those aspects of the software that are visible to the customer/user

Feedback is used to refine the prototype

(b) State two reasons why softwares fail . (2 marks)

Award one mark for each point

Increasing system complexity

Failure to use software engineering methods

(c) Explain the term Requirement Elicitation in software engineering. (2 marks

Correct answer award full mark

Requirements elicitation is an iterative process that can be represented as a spiral of activities; requirements discovery, requirements classification and organization, requirements negotiation, and requirements documentation.

The aims of the requirements elicitation process are to understand the work that stakeholders do and how they might use a new system to help support that work.

During requirements elicitation, software engineers work with stakeholders to find out about the application domain, work activities, the services and system features that stakeholders want, the required performance of the system, hardware constraints, and so on.

(d) State two examples of the functional and non-functional requirements in a Library system that allows users to search articles, e-books and also download using a single interface to a number of databases of articles in different libraries. (4 marks)

Award two marks for any correct points

- The user shall be able to search either all of the initial set of databases or select a subset from it.
- The system shall provide appropriate viewers for the user to read documents in the document store
- Every order shall be allocated a unique identifier (ORDER_ID) which the user shall be able to copy to the account's permanent storage area.
- The system shall be available to users 24/7
- The appearance of the user interface should be appealing

(e) For a system software to be a good software, it must exhibit some characteristics. Outline four such characteristics **(4 Marks)**

Award a mark for each point.

- Efficiency

Software should not make wasteful use of system resources (disk and memory space, CPU time, etc.) and should present appropriate response times

- Usability (ease of use)

Software must be usable by the users for which it was designed

- Dependability (reliability, availability, security, safety,…)

Software must be trustworthy

- Maintainability (ease of maintenance)

Software must evolve to meet changing needs

Software costs more to maintain than it does to develop. For systems with a long life, maintenance costs may be several times development costs

QUESTION 2: (COMPULSORY) 15 MARKS

(a) As a software engineer in an IT firm, you are expected to portray software engineering ethics. List four examples of these engineering ethics. **(4 marks)**

Award each point two marks

Confidentiality:

You should normally respect the confidentiality of your employers or clients regardless of whether or not a formal confidentiality agreement has been signed. Competence:

You should not misrepresent your level of competence. You should not knowingly accept work that is outside your competence.

Intellectual property rights :

You should be aware of local laws governing the use of intellectual property such as patents and copyright. You should be careful to ensure that the intellectual property of employers and clients is protected.

b) Explain two kinds of software products. **(4 marks)**

Award two marks for each point

Custom - developed for a particular customer, according to its specifications

Generic ("package") - developed for a general market, to be sold to a range of different customers

c) Describe the importance of software Engineering **(3 marks)**

Award full mark

It is very important for businesses as it helps them distinguish from competitors and become more competitive. Software development can improve the client's experiences, bring more feature-rich and innovative products to market, and make setups more safe, productive, and efficient.

d) Explain two benefits of using the prototype model in software engineering. **(4 marks)**

- This model is more suitable where customer needs not cleared.
- This Type of model use in that area where the size of problem are large.
- It decreases the cost of maintenance. The main factor of using this model is its cost reductions.
- This model is very helpful in decrease the gap of communication between those persons who use the software and that group of developed the software system.
- In the comparison of final software alteration in prototyping model alteration is quickly and cheaper.

SECTION B (Carries 39 Marks)

QUESTION 3: (13 MARKS)

e) Kiptoo is a project manager in a water development project in Kilifi county. The project will install sensors to monitor the water pressure levels as well as detect leakages. The system will be an integrated system comprising water management aspect. Kiptoo and his team are developing the requirement specification document for the system. Agile software development method was adopted.

Required:

i. Highlight six quality components necessary for the documents of software specification.

(6 marks)

Award each point two marks

Completeness - The base of specified software system documents is to cover all points which satisfied the customer. In case if complete information not find in those document then a number of question raised against this and it will not be good for the developer and client. Both parties lose their trust from this. In short, a software satisfied documents describe all the points related to its function and performance.

Accurate - 100% accuracy should be present when a software specification document present because it misguide the client if this report is not accurate.

Understandable - Software specification document should not be confusing that means the vision or objective should be clear and understandable so that user can easily understand it and become satisfied with this report.

Regularity - Regularity is also a main part of this document because every part or point covered step by step so that trust can be built in the favor of these documents.

Alterable - Alteration quality also should be present in software specification document because need of client and software can be any time change. So for this change a large scope should be present in this document report.

Followable - If the software specified document covered all point step by step then any one can easily understand it and checked it again at every level. In short cross check is easily done with this report.

Valid - All document related to it should be valid by law, requirement and the client need.

ii. Explain two benefits of using the above methodology in software development. **(4 marks)**

Award each point one mark.

- Allow the software and the software process to be quantified
- Measures of the software process or product
- May be used to predict product attributes or to control the software process

iii. State the reason for validation in software development process . **(3 marks)**

This is process of deciding that the valuation of the software at the end of the process is as per the actual need.

Validation includes actual test results and use of software when verification process gets completed.

Validation decides that developed product meets required needs.

Validation insures that software being developed fulfills functional as well as non-functional needs.

QUESTION 4: (13 MARKS)

a) Explain the purpose of case tools. **(1 marks)**

CASE is the use of IT in SW development activities, techniques and methodology

CASE tools are programs that automate or support one or more phases in a SW development life cycle

- increase the speed of SW development activities
- increase the SW productivity
- improve the quality of the SW developed

b) With the aid of a diagram discuss six CASE repository functions . **(12 marks)**

Data Integrity

include functions to validate entries to the repository

ensure consistency among related objects

automatically perform 'cascading' modifications when a change to one object demands some change to objects that are related to it

Information sharing

provides a mechanism for sharing information among multiple developers and between multiple tools, managers and controls multi-user access to data and locks/unlocks objects so that changes are not overlaid on one another.

Data-tool integration

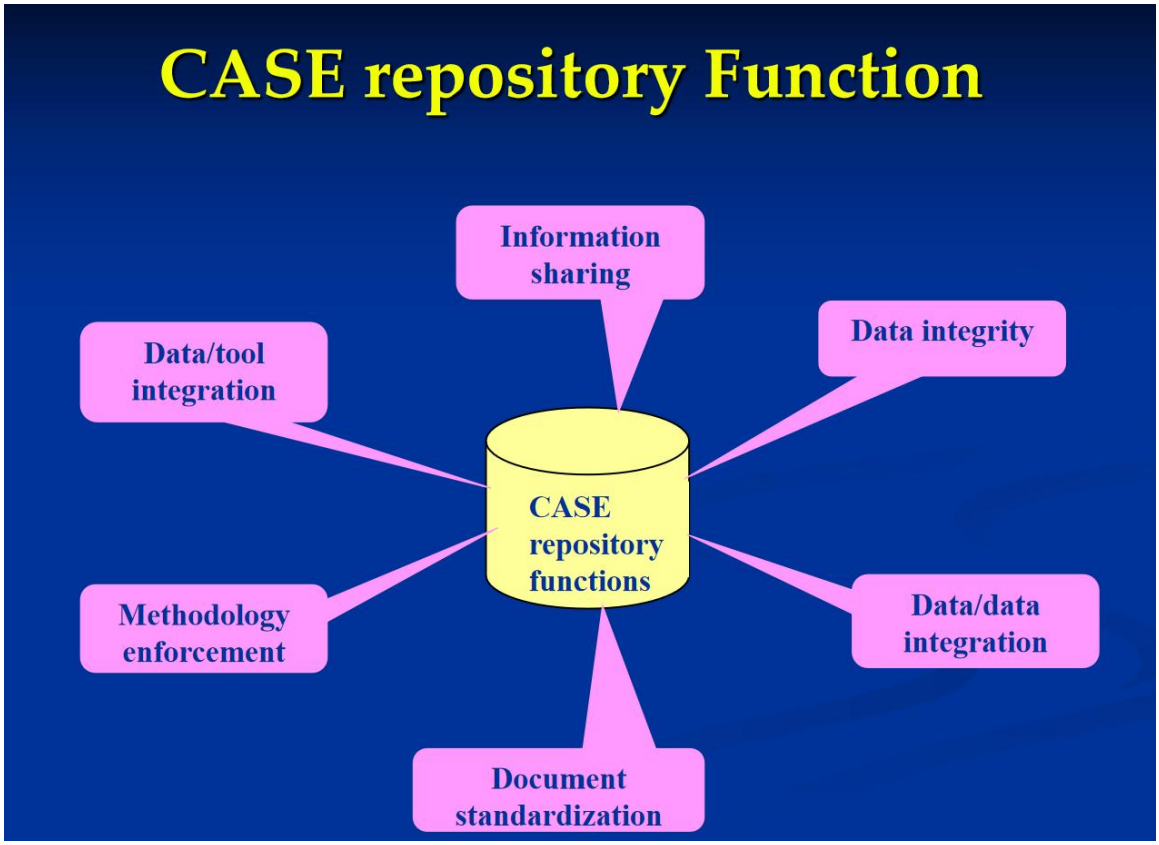
establish a data models that can be accessed by all tools in the I-Case environment, control access to that data, and perform appropriate configuration management functions.

Data Integration

the database management system relates data objects so that other function can be achieved.

Methodology enforcement

the ERD model of data stored in the repository can imply a specific paradigm for SE at a minimum , the relationships and objects define a set of steps that must be conducted to build the contents of the repository.



Document standardization

the definition of objects in the database leads directly to a standard approach for the creation of SE documents.

QUESTION 5: (13 MARKS)

a) Deep Tech company is a software development company that develops softwares to aid business in automating their processes. They have adopted the Object Oriented Design(OOD) approach in their projects.

Required

i. Explain the object oriented design approach. **(1 marks)**

Award two marks

Object oriented design works around the entities and their characteristics instead of functions involved in the software system. This design strategies focuses on entities and its characteristics. The whole concept of software solution revolves around the engaged entities

ii. Discuss three components of object oriented design. **(6 marks)**

Award each component a mark

- Objects - All entities involved in the solution design
- Classes - A class is a generalized description of an object.
- Encapsulation - In OOD, the attributes (data variables) and methods (operation on the data) are bundled together is called encapsulation.
- Inheritance - OOD allows similar classes to stack up in hierarchical manner where the lower or sub-classes can import, implement and re-use allowed variables and methods from their immediate super classes.
- Polymorphism - OOD languages provide a mechanism where methods performing similar tasks but vary in arguments, can be assigned same name.

iii. Highlight three unified modeling language(UML) suitable for the above design.

(6 marks)

Class Diagram
State Diagram
Activity Diagram
Component Diagram

QUESTION 6: (13 MARKS)

a) Describe test cases in software engineering. **(1 mark)**

A test case may be defined as a set of instructions for getting an error in system by causing a failure. Testing software is not so much expensive in the comparison of software testing.

b) Discuss Six Software Metrics . **(12 marks)**

Software metric	Description
Fan in/Fan-out	Fan-in is a measure of the number of functions or methods that call some other function or method (say X). Fan-out is the number of functions that are called by function X. A high value for fan-in means that X is tightly coupled to the rest of the design and changes to X will have extensive knock-on effects. A high value for fan-out suggests that the overall complexity of X may be high because of the complexity of the control logic needed to coordinate the called components.
Length of code	This is a measure of the size of a program. Generally, the larger the size of the code of a component, the more complex and error-prone that component is likely to be. Length of code has been shown to be one of the most reliable metrics for predicting error-proneness in components.
Cyclomatic complexity	This is a measure of the control complexity of a program. This control complexity may be related to program understandability. I discuss how to compute cyclomatic complexity in Chapter 22.
Length of identifiers	This is a measure of the average length of distinct identifiers in a program. The longer the identifiers, the more likely they are to be meaningful and hence the more understandable the program.
Depth of conditional nesting	This is a measure of the depth of nesting of if-statements in a program. Deeply nested if statements are hard to understand and are potentially error-prone.
Fog index	This is a measure of the average length of words and sentences in documents. The higher the value for the Fog index, the more difficult the document is to understand.

QUESTION 7: (13 MARKS)

Safaricom PLC hired you as software engineer to be part of a project team that will develop a software for use by the blind to access online materials such as journals, e-books and other useful material for their study.

a) Given the above information, required:

i. Highlight six types of risks that may be encountered during the project **(6 marks)**

Technology risks-

The database used in the system cannot process as many transactions per second as expected. Software components which should be reused contain defects which limit their functionality.

People risks-It is impossible to recruit staff with the skills required. Key staff are ill and unavailable at critical times. Required training for staff is not available.

Organisational risks-The organisation is restructured so that different management are responsible for the project. Organisational financial problems force reductions in the project budget.
Tools risks-The code generated by CASE tools is inefficient. CASE tools cannot be integrated
Requirements-Changes to requirements which require major design rework are proposed.
Customers fail to understand the impact of requirements changes.
Estimation risks-The time required to develop the software is underestimated. The rate of defect repair is underestimated. The size of the software is underestimated.

ii. Explain how hardware and software reliability are related. **(3 marks)**

The ability of a software or component to perform its required functions under stated conditions for a specified period of time.

Reliability of a software can be defined as a software which have no failure and working in a special time period with special environment.
Working of software depends on instructions while hardware is a part of mechanical.
Hardware and software both have major role in functional area of software. Because as we operate the software with hardware parts then if we find any errors in that case it effects the whole operation.
Reliability of software is maintained until any fault find in hardware which affects the path of the data.
Reliability of a software refers to the probability of a failing hardware part and the calculation of time for the maintenance of hardware parts. Physically errors always find in software system.

iii. Explain two stakeholders involved in the project. **(4 marks)**

Customer
Software Engineers
Software Developers
Users